Chapter 4: Continuous Random Variables and Probability Distributions

PART D:

- NORMAL APPROXIMATION TO BINOMIAL
- NORMAL APPROXIMATION TO POISSON

Learning Objectives:

At the end of this chapter, the students would be able to:

- ✓ Understand the concept of a continuous random variable and probability distribution
- ✓ Be able to compute the expected value, variance, and standard deviation for a continuous random variable
- ✓ Be able to compute probabilities using a Uniform distribution, Normal distribution, Normal approximation to the Binomial distribution and Normal approximation to the Poisson distribution
- ✓ Continuous Probability Distributions (Probability Mass Functions, Mathematical Expectations)
- ✓ Special Continuous Probability Distributions (Uniform Normal; Normal Approximation to the Binomial Distribution; Normal Approximation to the Poisson Distribution)

Continuity Correction Factor

Formula(s):

$$P(X < a) = P(X < a - 0.5)$$
$$P(X \leq a) = P(X < a + 0.5)$$
$$P(X > a) = P(X > a + 0.5)$$
$$P(X \geq a) = P(X > a - 0.5)$$
$$P(a < X < b) = P(a + 0.5 < X < b - 0.5)$$
$$P(a \leq X \leq b) = P(a - 0.5 < X < b + 0.5)$$

Continuity Correction Factor

Example 1:

$$P(X \geq 50)$$

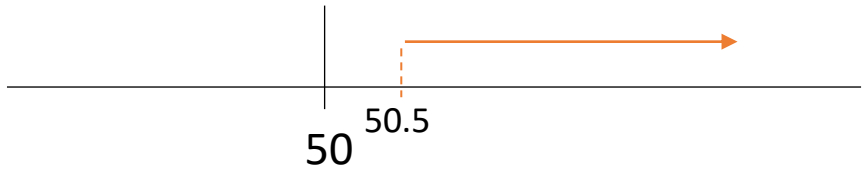


Draw an arrow going to the right, touching the 50

Continuity Correction Factor

Example 2:

$$P(X > 50)$$



Draw an arrow going to the right, without touching the 50

Continuity Correction Factor

Exercises:

i) $P(X \leq 50)$

ii) $P(X < 50)$

iii) $P(45 \leq X \leq 50)$

iv) $P(45 < X < 50)$

4.3 Normal Approximation to Binomial Distribution

Discrete

Binomial

$X \sim bin(n, p)$

$n \geq 30$
$n \times p \geq 5$
and
$n \times q \geq 5$

Continuous

Normal

$X \sim N(\mu = np, \sigma^2 = npq)$

Changing discrete distribution to continuous random distribution requires **continuity correction factor (+0.5) or (-0.5)

4.3 Normal Approximation to Binomial Distribution

Example 3 (from textbook Introduction to Probability and Statistics, UUM, 2021); pg. 99)

Suppose a fair coin is tossed 100 times. What is the probability there will be more than 65 heads? (ANS: 0.00097)

4.3 Normal Approximation to Binomial Distribution

Example 4:

40% of UiTM students in the main campus were not provided a hostel accommodation. If 100 students are selected at random, find the probability that more than 45 students are not provided hostel accommodation. (ANS: 0.1314)

4.3 Normal Approximation to Binomial Distribution

Discrete

Poisson

$X \sim \text{Poi}(\lambda)$

$\lambda \geq 20$

Continuous

Normal

$X \sim N(\mu = \lambda, \sigma^2 = \lambda)$

Changing discrete distribution to continuous random distribution requires **continuity correction factor (+0.5) or (-0.5)

4.3 Normal Approximation to Poisson Distribution

Example 5 (from textbook Introduction to Probability and Statistics, UUM, 2021); pg. 100)

If the queuing is 1.8 arrivals per minutes, find the probability of more than 70 arrivals in half an hour? (ANS: 0.01222)

4.3 Normal Approximation to Poisson Distribution

Example 6

Suppose the number of babies born during a day shift at a hospital follows a Poisson distribution with a mean of 6 per hour. What is the probability that during a particular 4-hour period, less than 20 babies are born? (ANS: 0.1788)